List of Published Scientific Works on the Dissertation Topic

Roger Nick Anaedevha (A23 – 501)

PhD candidate, Department of Cybernetics No. 22,
Institute of Intelligent Cybernetic Systems (IICS)
National Research Nuclear University "MEPhI", Moscow, Russia

Dissertation title: "Development of Adversarial Resilient Models Based on Artificial Intelligence in Hybrid Intrusion Detection and Prevention Systems for Network Security"

ORCID: 0000-0003-3649-1561 · Google Scholar · Github.com/rogerpanel/
ar006@campus.mephi.ru · rogernickanaedevha@gmail.com

Notation: Defended Provisions

In accordance with the section "Main scientific results submitted for defence" of the dissertation:

- **Provision No. 1** — Models M1 (CT-TGNN, SDE-TGNN) for certified dynamics of the learnable detector of the Hybrid IDPS (invariant P1).
- **Provision No. 2** — Algorithm M2 (FedLLM-API) for privacy-preserving distributed training of the Hybrid IDPS (invariant P4).
- **Provision No. 3** — Architecture M3 (TripleE-TGNN) for multi-level data representation in the Hybrid IDPS (invariant P6).
- **Provision No. 4** — Architecture M4 (MambaShield) for efficient inference of the Hybrid IDPS (invariant P3).
- **Provision No. 5** — Models M5–M6 (stochastic transformer and UC-HGP) for drift-aware uncertainty-calibrated adaptation of the Hybrid IDPS (invariants P5, P6).
- **Provision No. 6** — Unified Hybrid IDPS certification framework (M7) that integrates models M1–M6 through Stackelberg game-theoretic equilibria and is validated on network traffic (Chapter 5) and on UAV onboard systems (Chapter 6).

Note: this list includes **only published** (in-print) journal articles and conference proceedings on the dissertation topic. Preprints, manuscripts under review, and works accepted for publication but not yet in print at the time this list was compiled are deliberately excluded.

1. Published Journal Articles

1. **Anaedevha, R.N., Trofimov, A.G., Borodachev, Y.V. (2025).** "MambaShield: Selective State-Space Models for Poisoning-Resilient Sequential Modeling." *Expert Systems with Applications*, Elsevier. DOI: 10.1016/j.eswa.2026.131175. ISSN 0957-4174.

   Status: *published;* ***Indexing:*** *Scopus Q1, WoS Core (SCIE), VAK (K1), RSCI; JCR IF ≈ 7.5;* ***Defended provisions claimed:*** *No. 4 (M4 – MambaShield).*

2. **Anaedevha, R.N., Trofimov, A.G., Borodachev, Y.V. (2026).** "Uncertainty-calibrated hierarchical Gaussian processes for intrusion detection with multi-scale temporal modeling." *Neurocomputing*, Elsevier, p. 133105. DOI: 10.1016/j.neucom.2026.133105. ISSN 0925-2312.

   Status: *published;* ***Indexing:*** *Scopus Q1, WoS Core (SCIE), VAK (K1), RSCI; JCR IF ≈ 5.5;* ***Defended provisions claimed:*** *No. 5 (M5–M6 – stochastic transformer and UC-HGP).*

3. **Anaedevha, R.N., Trofimov, A.G., Borodachev, Y.V. (2026).** "Temporal Adaptive Neural Ordinary Differential Equations with Deep Spatio-Temporal Point Processes for Real-Time Network Intrusion Detection." *Complex and Intelligent Systems*, Springer. DOI: 10.1007/s40747-026-02291-7. ISSN 2199-4536.

   Status: *published;* ***Indexing:*** *Scopus Q1, WoS Core (SCIE), VAK (K1), RSCI; JCR IF≈5.0;* ***Defended provisions claimed:*** *No. 1 (M1 – CT-TGNN, SDE-TGNN).*

4. **Anaedevha, R.N., Trofimov, A.G. (2024).** "Improved Robust Adversarial Model against Evasion Attacks on Intrusion Detection Systems." *Optical Memory and Neural Networks (Information Optics)*, Vol. 33, Suppl. 3, pp. S414–S423, Springer. DOI: 10.3103/S1060992X24700681. ISSN 1060-992X.

   Status: *published;* ***Indexing:*** *Scopus (Q3, Computer Science misc.), RSCI;* ***Defended provisions claimed:*** *No. 6 (M7 – unified certification framework, foundational adversarial-robustness formulation).*

2. Published Conference Proceedings

5. **Anaedevha, R.N., Trofimov, A.G. (2025).** "Stochastic Multimodal Transformer with Uncertainty Quantification for Robust Network Intrusion Detection." In: *Advances in Neural Computation, Machine Learning, and Cognitive Research IX. NEUROINFORMATICS 2025.* Studies in Computational Intelligence, vol. 1241, Springer, Cham. DOI: 10.1007/978-3-032-07690-8_35. ISBN 978-3-032-07689-2.

   Status: *published;* ***Indexing:*** *Scopus (Springer SCI), RSCI;* ***Defended provisions claimed:*** *No. 5 (M5 – stochastic multimodal transformer with uncertainty quantification).*

6. **Anaedevha, R.N., Trofimov, A.G. (2025).** "Integrated Deep Q-Networks and Actor-Critic Models for Enhanced Poisoning Attack Detection in Network Intrusion Detection Systems." In: *2025 Conference of Young Researchers in Electrical and Electronic Engineering (ElCon)*, ETU-LETI, Saint-Petersburg, Russia. IEEE Xplore, vol. 33, suppl. 3, pp. 556–567.

   Status: *published;* ***Indexing:*** *IEEE Xplore, Scopus, RSCI;* ***Defended provisions claimed:*** *No. 6 (M7 – game-theoretic and reinforcement-learning response component).*

7. **Anaedevha, R.N. (2024).** "Application of Machine Learning Models for Device Identification in Wireless Network Traffic." In: *2024 Conference of Young Researchers in Electrical and Electronic Engineering (ElCon)*, Saint-Petersburg, Russia. IEEE Xplore, pp. 104–110. DOI: 10.1109/ElCon61730.2024.10468413.

   Status: *published;* ***Indexing:*** *IEEE Xplore, Scopus, RSCI;* ***Defended provisions claimed:*** *No. 1 (M1 – foundational feature-extraction problem statement for the NIDPS detector).*

3. Summary Counts by Type and Level

Total published scientific works on the dissertation topic: 7, of which:

Category	**Count**	**Share of 7**
By type of edition		
Journal articles	4	57.1%

Category	Count	Share of 7
Conference proceedings (full text, indexed in Scopus / IEEE Xplore)	3	42.9%
By indexing (all 7 published works)		
Scopus Q1, WoS Core (SCIE), VAK (K1) — journal articles	3	42.9%
Scopus (Q3, Computer Science misc.), RSCI — journal article	1	14.3%
Scopus (Springer SCI), RSCI — conference series	1	14.3%
IEEE Xplore (Scopus), RSCI — conference proceedings	2	28.6%
By authorship		
First author	7	100.0%
By Higher Attestation Commission (VAK) requirements (minimum 3 for engineering)		
Journal articles in editions on the VAK list and in Scopus/WoS	4	—
Of those, in editions of VAK category K1 and Scopus Q1	3	—

Compliance with the Regulation on the Award of Academic Degrees (Government of the Russian Federation Decree No. 842 of 24.09.2013, paragraphs 11–13): the requirement on the minimum number of publications in peer-reviewed scientific editions (not less than 3 for engineering sciences) is **fulfilled and exceeded**: 4 published journal articles, of which 3 are in Q1 Scopus / WoS Core / VAK K1 editions.

4. Cross-Reference: Publications vs. Defended Provisions

Provision	Published works	Additional confirmation
No. 1 (M1: CT-TGNN, SDE-TGNN)	[3] Anaedevha, R.N. *et al.*, *Complex & Intelligent Systems*, Springer, 2026 (Q1) [7] Anaedevha, R.N., ElCon 2024, IEEE Xplore (foundational)	Ksyrix LLC act: detection of multi-stage attacks on the network-interaction graph
No. 2 (M2: FedLLM-API)	*(no published works on this provision at the time of compiling this list; supported indirectly via active platform use — see Section 5)*	NII IKT LLC act: active use of the RobustIDPS.ai platform, which includes the FedLLM-API module, is explicitly confirmed

Provision	Published works	Additional confirmation
No. 3 (M3: TripleE-TGNN)	*(no published works on this provision at the time of compiling this list; supported by implementation acts — see Section 5)*	Ksyrix LLC act: identification of coordinated anomalies and multi-level representation of network interactions; AI Center, NRNU MEPhI act (in formalisation)
No. 4 (M4: MambaShield)	[1] Anaedevha, R.N. *et al.*, *Expert Systems with Applications*, Elsevier, 2025 (Q1)	NII IKT LLC act: integration of poisoning-resilient network-traffic anomaly-detection models into the operating IDS of the organisation
No. 5 (M5–M6: stochastic transformer, UC-HGP)	[2] Anaedevha, R.N. *et al.*, *Neurocomputing*, Elsevier, 2026 (Q1) [5] Anaedevha, R.N., Trofimov, A.G., NEUROINFORMATICS 2025, Springer SCI	Ksyrix LLC act: uncertainty-based identification of previously unknown patterns and adaptation to changes in traffic statistics
No. 6 (M7: unified certification framework, game-theoretic)	[4] Anaedevha, R.N., Trofimov, A.G., OMNN, Springer, 2024 (foundational adversarial robustness) [6] Anaedevha, R.N., Trofimov, A.G., ElCon 2025, IEEE Xplore (RL response)	NII IKT LLC act: joint operation of standard IDS algorithms and the proposed models within a unified system

Conclusion: four of the six defended provisions (No. 1, No. 4, No. 5, No. 6) are directly supported by published works; provisions No. 2 (M2 FedLLM-API) and No. 3 (M3 TripleE-TGNN) are, at the time of compiling this list, supported empirically — by the two received implementation acts (NII IKT LLC, Novosibirsk, and Ksyrix LLC, Moscow) and by the forthcoming act of the Artificial Intelligence Center of NRNU MEPhI, Moscow (see Section 5). When both forms of corroboration are combined (publications and implementation acts), coverage of all six defended provisions reaches **100 %**.

5. Implementation and Usage Acts

Two implementation acts have been received that confirm the practical applicability of the developed adversarial resilient models based on artificial intelligence in hybrid intrusion detection and prevention systems for network security; a third act is in the course of formalisation:

1. **Implementation act from *NII IKT LLC*** (Research Institute of Information and Communication Technologies), Novosibirsk (INN 5406600336, OGRN 1165476054567).
Issuance: approved by the General Director, Cand. Sc. (Eng.), Associate Professor E. A. Basynya on 03 June 2026; commission: Expert No. 1 — Head of Software Engineering Department, Cand. Sc. (Eng.), Associate Professor D. S. Khudyakov; Expert No. 2 — Head of Design and Construction Department N. A. Malyshkin.
Subject of implementation: integration of the developed poisoning-resilient network-traffic anomaly-detection models into the operating intrusion-detection system (IDS) of the organisation; active use of the RobustIDPS.ai software platform by in-house cybersecurity specialists for experimental research on information-security protocols and for continuous network-traffic analysis aimed at preemptive identification of potential vulnerabilities.
Confirmed effect (verbatim from the act): "joint operation of the standard IDS algorithms and the models proposed by the author significantly improved the accuracy of identifying suspicious network activity while preserving minimal data-processing latency, thereby raising the information-security level of the protected objects and minimising the risk of protection bypass."
Provisions confirmed: No. 2 (indirectly, via active use of RobustIDPS.ai), No. 4 (poisoning resilience), No. 6 (integration into existing protection means).

2. **Implementation act from *Ksyrix LLC***, Moscow (INN 9724072687, KPP 772401001, OGRN 1227700098449).
Issuance: signed by the head of the organisation D. E. Kravtsov and the person responsible for implementation S. M. Kharkov, 2026.
Subject of implementation: deployment of the developed solutions within the company's network-traffic analysis and infrastructure-protection system; integration of the software implementation (including experimental protocols and trained models) into the existing software of the organisation.
Confirmed effect (verbatim from the act): the solutions "successfully ensure: detection of complex multi-stage attacks and coordinated anomalies on the network-interaction graph, identification of previously unknown anomalous-behaviour patterns via uncertainty analysis, and adaptation to changes in the organisation's network-traffic statistics"; the use of the developed models and algorithms "increased the resilience of the detection system to potentially dangerous external influences and improved the quality of incident detection in a changing environment."
Provisions confirmed: No. 1 (CT-TGNN/SDE-TGNN graph dynamics), No. 3 (multi-level graph representation), No. 5 (uncertainty estimation and drift adaptation).

3. **Implementation act from the *Artificial Intelligence Center of the National Research Nuclear University "MEPhI"***, Moscow — *in the course of formalisation.*
Planned subject of implementation: application of the developed AI-based Hybrid IDPS framework (models M1, M3, M5, M6, M7) to the protection of UAV onboard systems within the three-tier "onboard – droneport – cloud" architecture (corresponds to Chapter 6 of the dissertation); confirmation of domain independence of the framework via transfer from the network-traffic sector (NIDPS) to the adjacent UAV onboard-system sector.
Provisions to be confirmed (planned): No. 1, No. 3, No. 5, No. 6.

Outcome: the two received acts together with the forthcoming third act collectively confirm *all* six defended provisions.

6. Software Products and Intellectual-Property Registration

1. **Software platform "RobustIDPS.ai"** — a unified open-source software platform that integrates 17 neural-network models into 10 functional groups and 51 pages of a Progressive Web Application user interface; implements all models M1–M7 as components of the unified Hybrid IDPS system, including a four-layer large-language-model defence framework (94.5 % detection across 517 attack scenarios) and the Multi-Agent PQC-IDS module.
   Source code deposit: Zenodo, DOI 10.5281/zenodo.19129512.
   Provisions confirmed: No. 1–No. 6 (all six, jointly within the unified system).

2. **Software-for-computers (SW) registration applications with Rospatent** are planned after dissertation defence for the key components of "RobustIDPS.ai" — CT-TGNN, FedLLM-API, TripleE-TGNN, MambaShield, and the unified M7 certification framework (a total of 5 applications).

3. **Invention patents:** none have been filed at the time of preparation of this list; applications are planned following production validation at Ksyrix LLC and NII IKT LLC for the Multi-Agent PQC-IDS and FedLLM-API components.

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Abbreviations: "K1", "K2" — categories of peer-reviewed scientific editions on the list of the Higher Attestation Commission (VAK) under the Ministry of Science and Higher Education of the Russian Federation; "Q1", "Q2" — journal quartiles in the Scopus / Web of Science databases at the time of manuscript submission; "IF" — two-year impact factor from Journal Citation Reports (JCR); "RSCI" — Russian Science Citation Index (RINTs).

Complete publication records and citation metrics are maintained on the author's Google Scholar and ORCID profiles.